1 **(a)** $6x - 15 - 2x - 8$ **[M1]**

$4x - 23$ **[A1]**

(b) $4xy(2x - 3y)$ or $4x(2xy - 3y^2)$ or $4y(2x^2 - 3xy)$ or $xy(8x - 12y)$ **[M1]**

$4xy(2x - 3y)$ **[A1]**

[Total Marks: 4]

2 **(a)** Correct expansion: $6x - 15 - 4x + 8$ **[M1]**

$2x - 7$ **[A1]**

(b) $6xy(2x - 3y)$ **[B2]**

Correct partial factorisation e.g., $6x(2xy - 3y^2)$ or $2xy(6x - 9y)$ **[M1]**

[Total Marks: 4]

3 **(a)** Expansion to give $6x - 15$ and $-4x + 12$ **[M1]**

$2x - 3$ **[A1]**

(b) $7xy(2x + 3y)$ **[B2]**

Partial factorisation, e.g. $7x(2xy + 3y^2)$ or $xy(14x + 21y)$ **[B1]**

[Total Marks: 4]

4 **(a)** $2x^2 + 10x - 3x - 15$ **[M1]**

$2x^2 + 7x - 15$ **[A1]**

(b) $c(a - 3b) + 2d(a - 3b)$ or $a(c + 2d) - 3b(c + 2d)$ **[M1]**

$(a - 3b)(c + 2d)$ **[A1]**

(c) $(4x - 5y)(4x + 5y)$ **[B1]**

[Total Marks: 5]

5 **(a)** $(x + a)(x + b)$ where $ab = 15$ or $a + b = -8$ **[M1]**

$(x - 3)(x - 5)$ **[A1]**

(b) $(2x + a)(x + b)$ where $ab = -12$ or $2b + a = -5$ **[M1]**

$(2x + 3)(x - 4)$ **[A1]**

(c) $a(x - 3) + 2b(x - 3)$ or $x(a + 2b) - 3(a + 2b)$ **[M1]**

$(a + 2b)(x - 3)$ **[A1]**

[Total Marks: 6]

6 **(a)** Taking out common factor to reach $2(9x^2 - 25y^2)$ **[M1]**

Factorising difference of squares to reach $(3x - 5y)(3x + 5y)$ **[M1]**

$2(3x - 5y)(3x + 5y)$ **[A1]**

(b) Taking out common factor x to reach $x(3x^2 + 10x - 8)$ **[M1]**

Factorising quadratic to reach $(3x - 2)(x + 4)$ **[M1]**

$x(3x - 2)(x + 4)$ **[A1]**

[Total Marks: 6]

7 (a) $(2a - b)(3x - y)$ **[B2]**

Correct partial factorisation e.g., $2a(3x - y) - b(3x - y)$ or $3x(2a - b) - y(2a - b)$ **[M1]**

(b) (i) Extracts common factor of 2: $2(25x^2 - 4y^2)$ **[M1]**

Factorises difference of two squares: $(5x - 2y)(5x + 2y)$ **[M1]**

$2(5x - 2y)(5x + 2y)$ **[A1]**

(ii) Extracts common factor of x: $x(2x^2 + 5x - 12)$ **[M1]**

Factorises quadratic: $(2x - 3)(x + 4)$ **[M1]**

$x(2x - 3)(x + 4)$ **[A1]**

[Total Marks: 8]

8 (a) Correct expansion of any two brackets, e.g., $(2x^2 - 5x - 3)$ or $(2x^2 + 9x + 4)$ or $(x^2 + x - 12)$ **[M1]**

Correct expansion of their quadratic with the third bracket, producing at least 5 terms, e.g., $2x^3 + 8x^2 - 5x^2 - 20x - 3x - 12$ **[M1]**

$2x^3 + 3x^2 - 23x - 12$ **[A1]**

(b) $(2x + 3)(x - 5)$ **[B2]**

Accept $(2x + a)(x + b)$ where $ab = -15$ or $2b + a = -7$ **[M1]**

[Total Marks: 5]

9 (a) $(x^2 + x - 6)(2x - 1)$ or $(x - 2)(2x^2 + 5x - 3)$ or $(x + 3)(2x^2 - 5x + 2)$ **[M1]**

$2x^3 - x^2 + 2x^2 - x - 12x + 6$ (allow one error in expansion) **[M1]**

$2x^3 + x^2 - 13x + 6$ **[A1]**

(b) $(3x \pm a)(2x \pm b)$ where $ab = 10$ or $6x^2 - 15x + 4x - 10$ **[M1]**

$(3x + 2)(2x - 5)$ **[A1]**

(c) $2x(x^2 - 9)$ **[M1]**

$(x - 3)(x + 3)$ seen **[M1]**

$2x(x - 3)(x + 3)$ **[A1]**

[Total Marks: 8]

10 **(a)** $8x - 12y - 3x + 15y$ **[M1]**

$5x + 3y$ **[A1]**

(b) $7ab$ extracted as a common factor **[M1]**

$7ab(2a - 3b)$ **[A1]**

[Total Marks: 4]

11 **(a)** $6x^2 + 2x - 15x - 5$ (at least 3 correct terms) **[M1]**

$6x^2 - 13x - 5$ **[A1]**

(b) $(x + a)(x + b)$ where $ab = -18$ or $a + b = -7$ **[M1]**

$(x - 9)(x + 2)$ **[A1]**

(c) $2(9x^2 - 25y^2)$ **[M1]**

$(3x - 5y)(3x + 5y)$ seen **[M1]**

$2(3x - 5y)(3x + 5y)$ **[A1]**

[Total Marks: 7]

12 **(a)** Expands any two brackets correctly, e.g., $(2x^2 - 5x - 12)(x + 1)$ or $(x - 4)(2x^2 + 5x + 3)$ or $(x^2 - 3x - 4)(2x + 3)$ **[M1]**

Multiplies their resulting quadratic by the third bracket, allowing one error **[M1]**

$2x^3 - 3x^2 - 17x - 12$ **[A1]**

(b) $(3x + a)(2x + b)$ or $(6x + c)(x + d)$ where $ab = -10$ or $cd = -10$ **[M1]**

$(3x + 2)(2x - 5)$ **[A1]**

(c) $x^2(x + 3) - 4(x + 3)$ or $x(x^2 - 4) + 3(x^2 - 4)$ **[M1]**

$(x^2 - 4)(x + 3)$ **[M1]**

$(x - 2)(x + 2)(x + 3)$ **[A1]**

[Total Marks: 8]